Page No.
Date

Q.1. Define Digital forensics. Explain its importance in investigation cyber crimes.

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Digital forensics is the branch of forensic science that deals with the identification, collection, analysis and preservation of digital evidence in a manner that is legly acceptable.

Importance:

- Digital forensic investigation helps in identifying and tracing cyber criminals through digital evidance
- It helps to collect and recover deleted or hidden data from computers and mobile device
- It helps investigator analyze how the cyber crime was somitted and reconstruct events.
- Digital forensics provides accurate and legally acceptable evidance for use in court.
- It helps organization detect security breaches and system misuse.
- Digital forensics also help in preventing future cyber crimes by improving security measures.

Scanned with OKEN Scanner

Page No.
Date

Q.2. Explain the rules of Digital forensics with suitable example:

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① Original evidence should not be changed:
The original digital data must be kept safe and should never be edited or opened directly
e.g.. If a laptop is seized, the investigator does not work on the original laptop but creates an exact copy and uses that copy for investigation.

② Integrity of evidence must be maintained:
The evidence should remain the same form the time it is collected until it is presented in court.
e.g.. Hash values are calculated before and after analysis to prove that the data was not modified

③ Proper documentation is necessary:
Every step taken during the investigation should be written clearly.
e.g.. The investigator record detail such as date, time, device type, and action performed during examination.

④ Only authorized and trained persons should handle evidence:
Digital evidence should be handled only by qualified forensic experts.
e.g.. A trained digital forensic officer analyze a

Scanned with OKEN Scanner

Page No.
Date

Seized mobile phone instead of an untrained
e.g.. person.

⑤ Legal rules must be followed:
Evidence should be collected according to legal procedures so it is valid in court
e.g.. police seize a computer only after getting proper legal permission or warrant.

⑥ chain of custody should be maintained.
A clear record should be kept of who handle the evidence and when.
e.g.. Every time evidence is transferred from one officer to another, it is recorded and signed.

Scanned with OKEN Scanner

Page No.
Date

Q.3. Describe the Digital Forensics Investigation process.

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① Identification:
In this step, the investigator first finds and identifies all digital devices that may contain evidence related to the cyber crime. This includes computer, mobile phones, memory card, hard disks, and network device.

② Preservation:
After identification, the digital evidence is preserved to protect it from damage, loss, or modification. The devices are keep safely and access is restricted so that no one can change the data.

③ Collection:
In this stage, the investigator collects the digital evidence using proper forensic metho Exact copies of the data are created so the original evidence remain untouched.

④ Examination:
The collected data is carefully examined to locate important information such as deleted files, emails, browsing history, logs, and suspicious activities.

Scanned with OKEN Scanner

Page No.
Date

(5) Analysis:
The examiner studies the examined data to understand the sequence of events and determine how the cyber crime was committed and who may be responsible.

(6) Presentation (Reporting):
finally, all finding are documented in clear and simple report. This report is presented in court or to authorities as valid digital evidence.

Q.4 Explain different types of Digital forensics:
① computer forensic ② Network forensics
③ Mobile forensic ④ cloud forensics
⑤ Data forensic.

→

① computer forensics:
computer forensics is used to investigate computer and laptop involved in cyber crimes. It helps in finding evidence such as documents, emails, internet history, system logs, and deleted files stored on hard disks or other storage devices.

② Network forensics:
Network forensics deals with analysis of network traffic to identify cyber attack and unauthorized access. It help investigators track by ~~stu~~ studying IP addresses, data packets, and network logs.

Scanned with OKEN Scanner

Page No.
Date

③ Mobile forensics :
Mobile forensics focuses on the examination of mobile phone and tablets. It helps in recovering call details, SMS, contacts, photos, video, location data, and application information related to the crime.

④ cloud forensics :
cloud forensics involves the investigation of data stored on cloud platform and online services. It helps in collecting and analyzing data from cloud servers, online storage, email and cloud-based applications.

⑤ Database forensics :
Database forensics is used to investigate database system to detect unauthorized access or data changes. It helps in finding deleted records, modified data and misuse of database information

Scanned with OKEN Scanner

Page No.
Date

Q.5. Explain the areas of application of computer forensics in:
① Public sector ② Private sector.

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① Public sector:

- Law enforcement:
computer forensics is used by police and investigating agencies to solve cyber crimes such as hacking, online frud, and identify theft by examining digital evidence from computers and mobile devices.

- Judicial system:
It helps court examine and verify digital evidence so that judgment can be given based on accurate and legally valid electronic data.

- Government Agencies:
Computer forensics is used to investigate data leakage, misuse of goverment computers, corruption cases, and unauthorized access to official records.

- Defense and National security:
It is used to detect and investigate cyber attack, cyber espionage, and cyber terrorism that may affect national security system.

Scanned with OKEN Scanner

Page No.
Date

② Private Sector.

- Corporate Investigation:
computer forensics helps organization investigate internal fraud, employee misconduct, and misuse of company computer systems.

- Financial Institutions:
Banks and financial organization use computer forensics to investigate online banking fraud, illegal transactions, and cyber attacks.

- Intellectual Property Protection
It help companies protect confidential data such as design, software, and business secret from theft or illegal copying.

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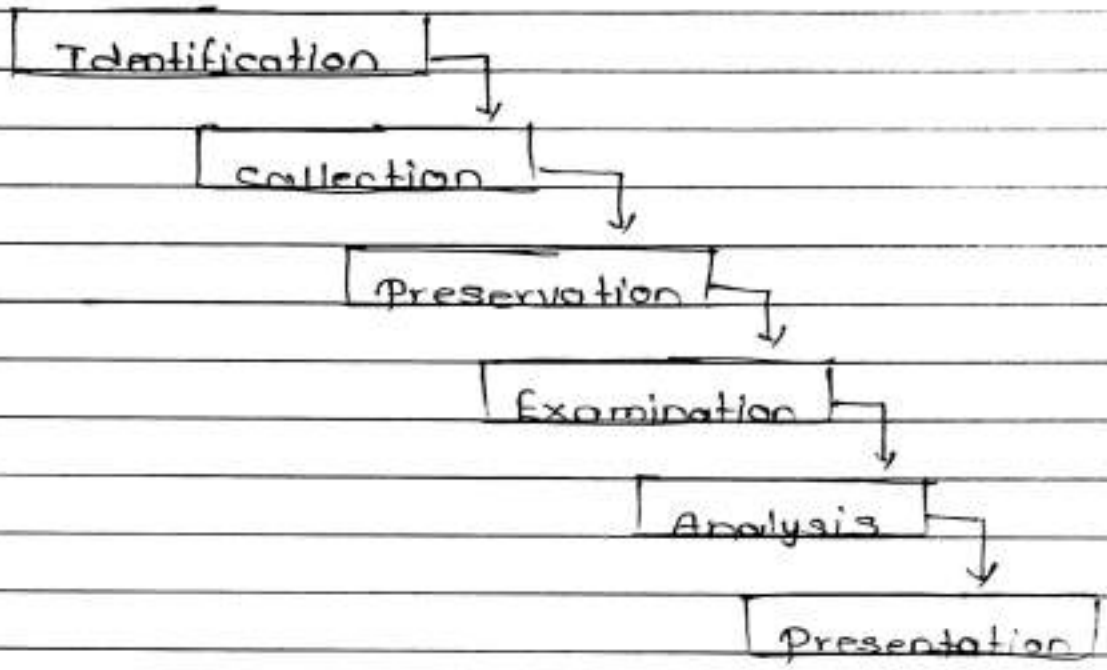
Q.6. Explain the Digital forensic process with neat diagram.




① Identifications:
In this step, the investigator identifies the digital devies and data related to the cyber crime, such as computer, mobile phones, hard disk, or servers.

② Preservation:
The identified digital evidence is preserved to protect it from damage, deletion, or modification. Proper care is taken so the data remains unchanged.

③ Collection:
Digital evidence is collected using forensic method. Exact copies of the data are created so that the original evidence is not disturbed.

Scanned with OKEN Scanner

Page No.
Date

④ Examination:
The collected data is examined carefully to find useful information like deleted files, email, logs, and browsing history.

⑤ Analysis:
In this step, the examiner analyzes the examined data to understand how the crime happed and to identify the suspect.

⑥ Presentation (Reporting):
Finally, all finding are documented in a clear report and presented in court or to authorities as legal evidence.

Q.7. State the goal and application of Digital forensic.

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Goals of Digital forensic:
- The main goal of digital forensics is to identify and collect digital evidence related to cyber crimes.
- It aims to preserve digital evidence safely so that it is not altered or damaged.
- Digital forensics helps to analyze digital data in a scientific and systematic manner.
- It ensures that digital evidence is legally acceptable and can be presented in court.
- It helps investigators understand how the cyber crime was committed and who was involved.

Scanned with OKEN Scanner

Page No.
Date

Applications of Digital forensics:

- Digital forensics is applied in cyber crime investigation such as hacking, online fraud, and identity theft.
- It is used by law enforcement agencies to collect and examine digital evidence in criminal cases.
- Digital forensics is used in corporate investigation to detect employee misconduct and data theft.
- It is applied in financial institution to investigate online banking and financial frauds.
- Digital forensics is used to recover deleted hidden, or damaged digital data from electronic devices.

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